

CPT6344 SOCIAL MEDIA COMPUTING 2026 ASSIGNMENT (50%)

ASSIGNMENT GUIDELINE

1. INTRODUCTION

- This assignment contributes 50% of your coursework marks. You will work collaboratively in groups of up to 3 students from the same Tutorial Section. The assignment is divided into three parts which must be completed:
 - Assignment Part 1 (30%):
 - **CLO 3:** Produce different types of professional and technical documents
 - **Deliverable:** Final Report
 - Assignment Part 2 (10%):
 - **CLO 3:** Display the attributes of effective professional communication
 - **Deliverable:** PowerPoint Final Presentation
 - Assignment Part 3 (10%):
 - **CLO 1:** Display the attributes of effective project code
 - **Deliverable:** Project Code Repository
- Fill in the information needed on the Assessment Form (page xx of this document). Please insert the form in this document and indicate the page number.

2. SUBMISSION REQUIREMENTS

- Write the report using Microsoft Word 365.
- You can choose the titles & Dataset which have been approved by the instructor.
- General Outline:
 - Font - Times New Roman, 12pt, Black
 - Alignment - Justify
 - Spacing - 1.5 lines
 - Header : CDS6344 - Social Media Computing
 - Footer : Project Title, Page Number
 - Margin : Left: 1.5 inch
 - : Right/Top/Bottom: 1.0 inch
 - : Minimum number of pages for the report is 10, excluding Appendixes
- Submission file should include:
 - Cover Page
 - Assessment form
 - Final Report
 - PowerPoint slides
 - Turnitin Similarity and AI report
- Your submission should be in **ONE** file only. Combine all files (cover page, assessment form, final report, Turnitin & AI report, and PowerPoint slides) into one file.

- A cover page which should have the following items:
 - **Subject:** CDS6344 Social Media Computing •
 - **Lecture Section:** TCXL
 - **Tutorial Section:** TTXL or TXXL
 - **Group Number/Name:** G0X/GNAME
 - **Submitted to:** Tutor's Name
 - **Assigned Project Title (prior approval):** (Refer to your tutorial lecturer for the approved title, problem statement and dataset)
 - **List of group members:** IDs and Names as registered with MMU
- Name your file as **TutorialSection_TutorName_GroupNumber_ASGN.pdf**. The submission should be done in soft copy to eBwise. Please refer to your tutorial lecturer for more information on the file submission to eBwise.
 - (Soft copy) Report and power point presentation submission date: 11:59 pm 26/06/2026 submit in ebWise.

Presentation: Schedule will be announced by the Tutor.

(Students are advised to bring the printed copy (one) of the report at the time of presentation)

- **Only ONE submission per group by the group leader. Group Leader, please take note of this.**

3. DUE DATE OF SUBMISSION OF ASSIGNMENT and POWERPOINT SLIDES

- Due date for submission of Assignment Part 1 and PowerPoint slides in softcopy is on **Week 13 (Friday, 26th June 2026 at 11.59PM)**. Late submission with penalty is allowed up to **Sunday, 28th June 2026 11:59 pm. After Sunday, 28th June 2026**, we will **NOT ACCEPT** any more submission for Assignment.

4. WHAT SHOULD BE DONE? THE SITUATION

- This project aims to develop an end-to-end Natural Language Processing (NLP) pipeline that leverages state-of-the-art techniques to extract, analyze, and visualize sentiments, opinions, and aspect-based sentiments from a large textual dataset. The system combines classical machine learning approaches with modern deep learning architectures, including transformer-based models, to deliver an actionable understanding of textual data.
- Dataset
The selected dataset must offer rich, real-world textual data suitable for both document-level and aspect-level sentiment analysis.
- Important topics:
 - I. **Sentiment Analysis:** Classify sentiments into categories such as *positive*,

- negative*, and *neutral* using machine learning classifiers (Logistic Regression, SVM, Random Forest) and deep learning models (LSTM, BERT, RoBERTa).
- II. **Opinion Mining:** Extract explicit and implicit opinions from text using dependency parsing and opinion lexicons and identify opinion targets.
 - III. **Aspect-Based Sentiment Analysis (ABSA):** Decompose reviews to identify sentiments towards specific product aspects (e.g., “battery life”, “screen quality”), using techniques like spaCy-based noun phrase extraction and fine-tuned transformer models like BERT-ABSA.
 - IV. **Model Development and Evaluation:** Train, validate, and compare performance of traditional ML algorithms with advanced DL/Transformer models. Use metrics such as accuracy, F1-score, precision, recall, and confusion matrices.
 - V. **Visualization and Insight Generation:** Create detailed visualizations including sentiment distributions, word clouds, aspect-wise sentiment maps, and opinion trend graphs using tools like Matplotlib, Seaborn, Plotly, and NLP-specific visualization tools like spaCy’s display.
- **Methodology**
 - **Data Preprocessing:** Text normalization, tokenization, stopword removal, lemmatization.
 - **Feature Engineering:** TF-IDF, word embeddings (GloVe, Word2Vec), contextual embeddings (BERT embeddings).
 - **Modeling:**
 - **Traditional ML Models:** Naïve Bayes, Logistic Regression, SVM.
 - **Deep Learning Models:** BiLSTM, CNN for text, GRU.
 - **Transformers:** BERT, DistilBERT, RoBERTa (fine-tuned for classification and ABSA tasks).
 - **Hyperparameter Tuning:** Grid Search, Random Search, and Bayesian optimization using cross-validation. (optional)
 - **Evaluation:** Use k-fold cross-validation and comprehensive benchmarking.
 - **Tools & Technologies**
 - **Languages:** Python
 - **Libraries/Frameworks:** NLTK, spaCy, scikit-learn, TensorFlow, PyTorch, Hugging Face Transformers
 - **Visualization:** Matplotlib, Seaborn, Plotly, WordCloud, Streamlit
 - **Others:** Jupyter Notebook, Google Colab, GitHub.
 - **Objectives**
 - To train students on how to collect and organize raw datasets which can be obtained through the usage of Twitter or other resources (GitHub, Kaggle, etc).
 - To develop a robust, scalable NLP pipeline capable of handling multiple sentiment and opinion analysis tasks.

- To inculcate a high teamwork spirit and to build strong communication skills among students.

5. ASSIGNMENT PART 1 (30%) - Final Report

- Your group is tasked with writing a **final report** based on the agree Title, problem statement and Dataset. The **final report** should cover detailed information about the project, which includes the following:
- Report Preparation
 - Write the report using Microsoft Word 365.
 - You can choose the titles & Dataset which have been approved by the instructor.
 - General Outline:
 - Font - Times New Roman, 12pt, Black
 - Alignment - Justify
 - Spacing - 1.5 lines
 - Header: CDS6344 - Social Media Computing
 - Footer: Project Title, Page Number
 - Margin: Left: 1.5 inch
 - Right/Top/Bottom: 1.0 inch
 - Minimum number of pages for the report is 10, excluding Appendixes

Acknowledgment Any compliments or appreciation notes.		
Table of Content (must include the following 10 subsections)		
Use proper sequence and numbering.	Example:	Allocated Marks(30)
1.: Introduction	pg 1	(3 marks)
1.1: Project Overview	pg 2	(1 mark)
2.: Problem Statement	pg 3	(2 marks)
3.: Literature Review	pg 4	(5 marks)
4.: Methodology	pg 7	(5 marks)
5.: Sentiment Analysis		(2 marks)
6.: Transformers / Deep Learning models		(5 marks)
7.: Result & Visualization		(2 marks)
8.: Discussion		(2 marks)
9.: Conclusion & Future Work		(2 marks)
10.: References		(1 mark)
Project Overview A brief description of your project title and the objectives		
Content Explanations are simple and easily understood. Information is carefully selected, organized and presented with use of appropriate visualization tools.		
Conclusion Description of problems faced during the completion of the project and how you overcome them. Lessons learned and knowledge gained & Future directions.		

References

List down the details of sources (links/websites/research papers) to obtain information. Make sure you use the latest ones (Any references published before the year 2000 are considered obsolete, unless it is a fundamental research) Use- Chicago Manual or IEEE

Example: <http://sun.com/marketing/statistics>

1. Bill Daley, *Computers Are Your Future 2006*, Prentice Hall, 2006, pg. 20-30

Appendix

Link to the Github/COLAB repository, and other important data/flowcharts/pictures/photos that may not be suitable to fit into your report. Like the screenshot of dataset recourse location, etc.

6. ASSIGNMENT PART 2 (10%) - PowerPoint Presentation

Description:

In this part of the assignment, your group will create a **PowerPoint presentation** based on the **final report** that your group has created for **Assignment Part 1**.

PowerPoint Presentation Requirements:

- **Duration:** The entire presentation should last **10-12 minutes**. There will be Q&A session. Ensure that each group member contributes to the presentation, and each member in the group must present.
- All group members must be present physically in Formal attire.
- **Visuals:** Use relevant **visuals** (e.g., images, diagrams, charts) that help explain or support the content of your presentation. Visuals should be clear, high-quality, and complement the information presented.
- **Text on Slides:** You can use bullet points or short phrases to summarize key ideas. Your spoken explanation will provide the details.
- **Font s Layout:** Ensure consistency in font style, color scheme, and slide design.

Presentation Delivery:

- **Mode of Presentation:** Presentation will be conducted starting from **Week 14**. Your **tutorial lecturer** will discuss the presentation schedules for all the groups within the specific tutorial section.

- **Group Participation:** All group members should contribute to the presentation. Each student is expected to contribute equally to the development and delivery of the presentation.
- **Clarity s Confidence:** Speak clearly, confidently, and at a moderate pace. Make sure you are engaging in the audience and not reading directly from the slides.
- **Engagement:** Maintain eye contact with the audience, avoid filler words (e.g., "um", "like"), and aim for natural, professional delivery.
- **Practice:** Rehearse the presentation together as a group to ensure smooth transitions between speakers and a consistent tone throughout.
- **Presentation Schedule:** Your tutorial lecturer will discuss further with your group on the presentation schedule for this assignment.

7. ASSIGNMENT PART 3 (10%) - Project Code Repository

- All the codes must be posted on a GitHub/COLAB repository to be analyzed and accessed by the instructor.
- Both Instructor and Tutor **must be added as a collaborator** for the repository.
- Add your data visualization screen shots
- Add a paragraph for future work.

8. PENALTY

- **Penalties** and the total of marks deducted from the finalized marks:
- **No cover page and assessment form:** Deduction of 2 marks
- **No task distribution:** Deduction of 2 mark
- **Plagiarism above 20%: 0 mark (Note: This applies to all the members in the groups involved) (Similarity and AI score must be under 20%)**
- **Late Submission up to Sunday, 29th June 2026:** Deduction of 10 marks per day per submission
- **Did not follow instruction for submission such as file format, the required submitted files, etc:** Deduction of 5 marks

G. Rubrics for Assignment Part 1 Assessment (Group) - 30%:

Criteria	4 - Excellent	3 - Good	2 - Satisfactory	1 - Needs Improvement	0- Missing
Structure s Organization	The report is well-organized with clear sections and headings, making it easy to navigate and follow.	The report is organized into clear sections, though some sections could be more clearly defined or structured.	The report is somewhat organized, but some sections lack clarity or do not follow a logical flow.	The report lacks clear organization, making it difficult to follow.	Missing
Clarity Accessibility, spelling / grammar	The writing is clear, simple, and easy to understand. It is appropriate for a non-technical audience, with explanations of terms and concepts. Not a single typo or grammar mistake	The writing is mostly clear, though some parts may be difficult for a non-technical audience to understand. Only less than 5 typos/grammar mistakes; they are not that noticeable.	Some sections of the report are unclear or too technical, making it harder to follow. It consists of some typos / grammar mistakes	The writing is unclear, overly technical, or difficult for a non-technical audience to understand. It consists of many typos /grammar mistakes	Missing
Content -Table of Content - All the subsections from point no.6	The report provides a comprehensive, clear and detailed descriptions of all four aspects	The report provides a basic description, but it lacks sufficient detail or does not fully explain two key aspects.	The report provides a basic description, but it lacks sufficient detail or does not fully explain three key aspects.	The report provides a basic description, but it lacks sufficient detail or does not fully explain all four key aspects.	Missing
Content - operation guidelines troubleshooting guides -FAQ -support information	The report provides a comprehensive, clear and detailed descriptions of all four aspects	The report provides a basic description, but it lacks sufficient detail or does not fully explain two key aspects.	The report provides a basic description, but it lacks sufficient detail or does not fully explain three key aspects.	The report provides a basic description, but it lacks sufficient detail or does not fully explain all four key aspects.	Missing
Visuals & Presentation	The report includes high-quality, relevant visuals that help clarify the explanation of the system. The report is well-formatted and professional.	The report includes visuals, but they may not be fully relevant or clear. The formatting is generally professional.	The report includes minimal visuals, and the formatting could be improved.	The report lacks visuals, or the visuals do not support the text effectively. The formatting is unprofessional.	Missing
Turnitin & AI similarity report	Similarity: < 5% AI: 0%	Similarity: < 10% AI: 0%	Similarity: < 15% AI: 0%	Similarity: < 20% AI: 0%	Similarity: >20% AI: >20%

G. Rubrics for Assignment Part 2 Assessment - 10%:

(Evaluation is based on material displayed and used during presentation)

Criteria	4 - Excellent	3 - Good	2 - Satisfactory	1 - Needs Improvement	0- Missing
Clarity, organization and visual appeal	Slides are well-organized, with a clear structure and logical flow. Information is presented clearly and effectively. Uses relevant and high-quality visuals such as movie posters, key scenes, or clips that effectively support and enhance the presentation.	Slides are mostly organized with a clear structure. Information is generally clear and understandable. Uses relevant visuals that support the presentation but may need minor improvements in quality or relevance.	Slides have some organizational issues or unclear information. Presentation flow is somewhat disjointed. Uses some visuals, but they may be of lower quality or not always relevant to the presentation content.	Slides are poorly organized with unclear or confusing information. Presentation flow is disjointed. Uses few or irrelevant visuals that do not support the presentation effectively.	No slides
Time Management	Presentation flows smoothly and adheres strictly to the 12- minute time limit	Presentation flows generally well with minor time management issues	Presentation has some flow issues or time management problems.	Presentation is disorganized or poorly timed.	No presentation

G. Rubrics for Assignment Part 3 Assessment - 10%:

Criteria	4 - Excellent	3 - Good	2 - Satisfactory	1 - Needs Improvement	0- Missing
Individual Contribution	Actively contributed to the creation of the presentation, providing clear, detailed, and meaningful content. Took initiative in preparing the slides and participated equally in the planning and presentation delivery.	Contributed to the creation of the presentation with adequate input. Played a significant role in the planning and presentation, though the level of contribution may have been less than others in the group.	Contributed to the presentation, but the input was minimal or limited to specific sections. Participation during planning and delivery was not fully balanced with the rest of the group.	Limited or no contribution to the creation of the presentation. The individual played a minimal role in planning or presenting, affecting the overall group effort.	No contribution
Clarity and Communication	Delivered the presentation clearly and confidently. Ideas were communicated in a professional manner with exceptional clarity, avoiding jargon and ensuring that the audience could easily understand complex topics.	Delivered the presentation well, with clear communication and only minor lapses in clarity. Ideas were generally well-organized, but some sections might have been harder to follow.	Delivered the presentation with some difficulty, and parts of the presentation were unclear or lacked logical flow. May have used jargon or failed to explain complex topics sufficiently.	Presentation lacked clarity, with many sections difficult for the audience to follow. The speaker may have relied heavily on reading from slides or lacked explanation of key points.	No presentation
Visuals & Presentation	The report includes high-quality, relevant visuals that help clarify the explanation of the system. The report is well-formatted and professional.	The report includes visuals, but they may not be fully relevant or clear. The formatting is generally professional.	The report includes minimal visuals, and the formatting could be improved.	The report lacks visuals, or the visuals do not support the text effectively. The formatting is unprofessional.	Missing
Engagement, Professionalism and Delivery	Maintained excellent eye contact, used engaging body language, and kept the audience's attention throughout. The presenter delivered the presentation with outstanding professionalism. The speaker demonstrated confidence, and control over their material, with smooth transitions and no filler words.	Maintained good eye contact, though there were minor lapses in engagement or delivery. The presenter delivered the presentation with good professionalism, adequate confidence transitions or filler words. The presenter handled questions effectively.	Eye contact was limited and engagement with the audience was minimal. The presenter showed some hesitation, used filler words (e.g., "um", "like"), or lacked confidence in some part of the presentation.	Did not maintain eye contact, or lack of confidence or engage with the audience. The presenter may have appeared unprepared, frequently hesitating or disengaged.	No presentation

Instructor (affiliation):

Dr. Mohammad Shadab Khan, Assistant Professor, Faculty of Computing and Informatics, Multimedia University, Cyberjaya, Malaysia.

Office: BR4014

Email: shadab.khan@mmu.edu.my

GitHub: [Shadab-Khaan \(Shadab\)](#)

Dr. Alobaedy Mustafa Muwafak Theab, Assistant Professor, Faculty of Computing and Informatics, Multimedia University, Cyberjaya, Malaysia.

Office: BR4008

Email: mustafa.alobaedy@mmu.edu.my

Dr. Haque Radiah, Senior Lecturer, Faculty of Computing and Informatics, Multimedia University, Cyberjaya, Malaysia.

Office: FCI_BR0023

Email: radiah.haque@mmu.edu.my